POWER PULSE

# Symmetrical (3-Phase) Faults Important Formulae

Quick Revision Concepts by BALAJI SIR

## 1 Fault current

1. $$I_f = \frac{V_{pf}}{Z_{th} + Z_f} = \frac{1}{Z_{th} + Z_f} \text{ pu}$$

*pre-fault voltage 1.0 pu*

2. $$I_f(\text{A}) = I_{base} \times I_{f,pu}, \qquad S_{sc} = \sqrt{3} V_L I_f = \frac{MVA_{base}}{X_{pu}}$$

3. $$I_{sc} = I_{rated} \times \frac{100}{\%X}$$

## 2 Machine behaviour

4. $$I'' = \frac{E''}{X''_d} > I' = \frac{E'}{X'_d} > I = \frac{E}{X_d}$$

*subtransient > transient > steady state*

5. $i(t) = \text{AC (decaying)} + \text{DC offset}; \quad \text{asymmetrical rms} \approx 1.6 \times \text{symmetrical}$

*maximum offset*

## 3 Z-bus method

6. $$I_f(k) = \frac{V_k(0)}{Z_{kk} + Z_f}, \qquad V_i(F) = V_i(0) - Z_{ik} I_f(k)$$

7. $$I_{ij} = \frac{V_i(F) - V_j(F)}{z_{ij}}$$

*post-fault current in a line*

## 4 Reactor and breaker

8. $X_{reactor} = X_{required} - X_{existing}$ (pu, on a common base)

*current-limiting reactor*

9. $\text{Breaking MVA} = \sqrt{3}\, V_{rated}\, I_b \qquad \text{Making current} \approx 2.55\, I_{sc,rms}$

10. Neglect loads, resistance and pre-fault currents (simple method); use sub-transient reactance